

Ventilation with a Mechanical Ventilator

Guideline Title: Ventilation with a Mechanical Ventilator

Guideline Number: GUID-3203-PR024

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 2/20/2020, 6/8/2021	Developed by: Air Care Team
Approved by: Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

CCCXXXII. **PURPOSE:** To provide adequate support of ventilation and oxygenation to patients with an advanced airway during transport

CCCXXXIII. **DEFINITIONS:**

- A. BVM – Bag valve mask
- B. RR – Respiratory Rate
- C. ARDS – Acute respiratory distress syndrome
- D. COPD – Chronic Obstructive Pulmonary Disease
- E. I:E - Inspiratory : Expiratory Ratio
- F. ABG – Arterial Blood Gas
- G. PIP – Peak Inspiratory Pressure
- H. PEEP – Positive End Expiratory Pressure
- I. ETCO₂ - End-tidal CO₂

CCCXXXIV. **DEPARTMENT GUIDELINE:**

A. **EQUIPMENT:**

- i. ReVel Ventilator
- ii. Disposable ventilator circuit
- iii. Compressed oxygen source yielding 50 PSI output
- iv. Charged internal battery for ventilator, power supply for charging and power source
- v. Bag-valve-mask with oxygen reservoir
- vi. ETCO₂ detector
- vii. Pulse oximeter
- viii. Miscellaneous ventilator circuit adjuncts. (e.g., disposable PEEP valve, HEPA filter, heat-moisture exchanger, flex connector, aerosol medication chamber)

B. **INDICATIONS:**

- i. Any patient requiring placement of an artificial airway to support ventilation and/or oxygenation.

C. **CONTRAINDICATIONS:**

- i. Mechanical ventilation in patients with suspected tension pneumothorax is contraindicated until the tension pneumothorax has been resolved.

D. GENERAL EXPECTATIONS

- i. Perform a ventilator check at the beginning of each shift to ensure there is not any mechanical or electronic failure.
- ii. For interfacility transports, the ventilator should be removed from the aircraft and taken to the patient's side in any patient with a known advanced airway or potential need for advanced airway.
- iii. For scene flights, the ventilator should be attached to the patient after loading in the aircraft as soon as practice and safe with regard to aircraft lifting operations.
- iv. For long distance flights, calculate the anticipated oxygen need.

E. PROCEDURES:

- i. Verify artificial airway patency and position. Apply ETCO₂ detection and pulse oximetry equipment to verify artificial airway placement and to evaluate effectiveness of current ventilation technique per *GUID-PR009 -End-Tidal CO₂ Monitoring*.
- ii. Prepare the BVM device for emergent use in case of a ventilator failure
- iii. Obtain calculated and non-calculated ventilator settings and deploy mechanical ventilation therapy to the patient
 - 1. Identify the MINUTE VENTILATION needs of the patient by multiplying the respiratory rate and the tidal volume.
- iv. Ensure the desired settings are reaching the patient
- v. Drive changes based on pulse oximetry and ETCO₂ data.

F. REVEL VENTILATOR:

- i. Push the On/Off button on the lower interface panel. Verify that the ventilator powers up, displays the ventilator model name (Revel) or part number (XXXXX-XXX) followed by Same Patient in the display window.
- ii. Once the ventilator is turned on, the window displays Same Patient. Turn the scroll knob to the right until New Patient displays on the window. Push the Select button to confirm.
- iii. To set for patient size (The window displays Patient ID) Scroll the knob to the right until Patient Size displays on the window. Push the Select button to confirm. Then scroll the knob until the desired patient size displays (Infant, Pediatric or Adult). Push the Select button to confirm.
- iv. Once the patient size is selected, an option for INTUBATED or NPPV will show, scroll the knob until the desired option displays for Intubate or Noninvasive positive-pressure ventilation (NPPV). Push the Select button to confirm.
- v. Select appropriate mode
- vi. Set trigger to Volume-cycled if required.

- vii. Select desired Tidal Volume (V_t). Patients without shock should typically be ventilated with a V_t of 6-8ml/kg based upon ideal body weight. Tidal Volume equals lung protection.
 - 1. IBW formula is based on patient's height.
 - a. **Males** = $50 + 2.3 [\text{height (inches)} - 60]$
 - b. **Females** = $45.5 + 2.3 [\text{height (inches)} - 60]$
- viii. Select appropriate Respiratory Rate (RR).
 - 1. 16-18 breaths per minute should be starting point unless clinical presentation dictates otherwise. (Metabolic Acidosis, DKA, etc.)
 - 2. Respiratory rate equals ventilation. Adjust RR to achieve pH goals.
- ix. Select desired FiO_2 if applicable.
 - 1. An FiO_2 of 0.21-1.00 may be selected. Consider current ABG, SpO_2 and anticipated altitude changes when selecting an FiO_2 for transport.
 - 2. Consider that PaO_2 's greater than 300 are associated with increased mortality in traumatic brain injury. An FiO_2 of 0.5 or less is always safe, as PaO_2 's will not exceed 300 at this setting. In general, use the lowest FiO_2 setting that maintains SpO_2 at 93-99%.
- x. If Positive End Expiratory Pressure (PEEP) is required, select the desired amount and set via the ventilator's PEEP function.
 - 1. PEEP is typically set within the range of 1-20 cm H_2O . (See *Clinical Pearls* section, below, for more information.)
 - 2. FiO_2 and PEEP together equal oxygenation. See FiO_2 /PEEP scale below for recommendations for FiO_2 and PEEP.
- xi. Plateau Pressure (pPlat) – In volume mode, perform an inspiratory hold.
 - 1. pPlat should be less than 30cm H_2O .
 - 2. If higher than 30mm H_2O
 - a. Decrease tidal volume to prevent potential barotrauma.
 - b. Consider lung compliance problems and possibly switch to pressure control.
- xii. Select desired Inspiratory Time (I_t).
 - 1. Adult patients should receive an I_t of 0.8-1.5 seconds.
 - 2. Pediatric patients should receive an I_t of 0.5-1.0 seconds.
 - 3. Clinical goal should be to maintain an I:E ratio of no less than 1:2.
- xiii. The ventilator may be equipped with a default 1:2 I:E ratio setting. This may be selected instead of setting an I_t . Caution should be used in patients with an obstructive lung

component.

- xiv. Set high pressure alarm at 50cm H₂O and low-pressure alarm at 10cm H₂O. Occlude patient end of ventilator circuit and test for leaks, correct function, and observe delivery of several breaths.
- xv. Attach patient to the ventilator and observe peak inspiratory pressure (PIP).
 - 1. Reset high pressure alarm to 10-15cm H₂O above the PIP.
 - 2. Reset low inspiratory pressure alarm to 10-15cm H₂O below the PIP.
- xvi. Make note of mean airway pressure (P_{aw})
- xvii. Set low minute volume alarm if so equipped
- xviii. Observe delivery of several breaths. Evaluate patient for adequate chest rise, ETCO₂ and SpO₂. Adjust ventilator settings as necessary to improve clinical parameters.
 - 1. Monitor patient for AutoPEEP, cuff leaks, deadspace problems, and adequate sedation
- xix. Observe for alarms and immediately view alarm message to assist in identifying possible cause. (See *Alarm Troubleshooting* section, below, for more information.)
- xx. If at any time the ventilator should fail or an alarm is received that cannot be corrected, the patient should be immediately ventilated with a Bag-valve device attached to a 100% oxygen source until ventilator issue can be resolved.
- xxi. At the completion of the transport, dispose of ventilator circuit and clean ventilator as necessary

G. Alarm Troubleshooting:

- i. Crashing ventilator patient mnemonic to help troubleshoot problem.
 - 1. D - Dislodgement - Check ETCO₂, tube depth and listen for lung sounds
 - 2. O - Obstruction – Secretions, foreign object
 - 3. P - Pneumothorax – High PIP, decreased exhaled tidal volume, tachycardia, hypotension. JVD, trachea deviation late signs.
 - 4. E - Equipment – this can cause High and Low pressure alarms. Start at patient and work back to ventilator.
 - 5. S - Stack Breaths – Immediately disconnect vent circuit from ET tube. Often seen in COPD and asthma patient who have difficulty exhaling.
- ii. Possible causes of Low Inspiratory Pressure Alarm:
 - 1. Disconnect of one of the ventilator circuit tubes
 - 2. Decrease in source gas pressure (e.g., oxygen tank has dropped below 500psi).

3. Increase in lung compliance causing lowered Peak Inspiratory Pressure (PIP)
 4. Cuff leak
 5. Exhalation valve failure causing gas escape during inspiration
 6. Patient is spontaneously inspiring and requiring a greater flow than the ventilator is delivering.
- iii. Possible causes of High-Pressure Limit Alarm:
1. Occlusion of artificial airway
 - a. Kinked tube, mucous plug, or patient is biting on airway tube
 2. Decrease in lung compliance
 - a. Worsening pulmonary injury, bronchospasm, development of a pneumothorax, etc.
 3. Patient is coughing
 4. Exhalation valve stuck closed, not allowing patient to exhale adequately.

CCCXXXV. **DOCUMENTATION**

A. **EMS Charts**

- i. Record all set parameters, PIP and Paw trends in the patient transport record. Note in patient transport record that all alarms were set.

CCCXXXVI. **REFERENCES:**

- A. ARDSnet. (n.d.). Mechanical Ventilation Protocol Summary. Retrieved from http://www.ardsnet.org/files/ventilator_protocol_2008-07.pdf
- B. ARDSnet FiO₂/PEEP scale

Lower PEEP/higher FiO₂

FiO₂	0.3	0.4	0.4	0.5	0.5	0.6	0.7	0.7
PEEP	5	5	8	8	10	10	10	12

FiO₂	0.7	0.8	0.9	0.9	0.9	1.0
PEEP	14	14	14	16	18	18-24

Higher PEEP/lower FiO₂

FiO₂	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.5
PEEP	5	8	10	12	14	14	16	16

FiO₂	0.5	0.5-0.8	0.8	0.9	1.0	1.0
PEEP	18	20	22	22	22	24

TABLE 7.3 Prompted Changes based on Ongoing Physiological Data

Ongoing Ventilator Monitor		
Parameter	Level	Corrective Action:
Blood Pressure	Low	1. Fluid or pressor therapy, consider tension pneumothorax 2. Check for overbreathing 3. Consider "Resuscitation Mode"
SpO ₂	Low	↑ PEEP or FIO ₂
ETCO ₂	Low	↓ RR, check perfusion status (see Low Blood Pressure above)
	High	↑ RR
Frequency	High	Consider sedation; Patient is over-breathing the set ventilator rate
Exhaled Tidal Volume	Low	1. R/O air trapping (AutoPEEP of 0 confirms no air-trapping) 2. R/O small leaks - check for full tracheal tube pilot bulb 3. Assume a dead space problem: ↑ Vt (WHILE maintaining a PIP < 35 cm H ₂ O)
	High	N/A
Peak Inspiratory Pressure	High	1. Assess for plateau pressure 2. Rule out DOPES pneumonic • iE Dislodgement: tracheal tube misplaced • iE Obstruction: kink in tracheal tube, stepping on circuit, secretions • iE Pneumothorax iE Equipment: ensure tight and proper connections • iE Sedation: be sure to adequately sedate your patients High PIP/ Normal plateau pressure = Airway resistance problem; consider bronchodilators
	Low	1. Reconnect all connections (most likely the circuit is disconnected) 2. Increase Vt or PC (possibly too low)
Plateau Pressure	High	Inspiratory Hold Maneuver: HIGH PIP/ HIGH Plateau Pressure = Lung compliance problem; change to PC.
AutoPEEP	High	Expiratory Hold Maneuver: if AutoPEEP is > 0, the patient is air trapping; provide a longer E time by adjusting the I:E ratio to 1:4, 1:5, or 1:6. Also consider a bronchodilator to allow for more complete exhalation